

CS & AI • EXPLAINED

How does the internet reach your phone?

Spoiler: it's almost never "wireless".

THE HOOK

You tap a video in **India**. It loads from a server in **America** — in **milliseconds**.

No wire touches your hand. So... how?

Let's follow your data on its journey across the planet. 🌍



TODAY'S QUESTION

When you open a website,
what **physically** travels — and
how **far**?

IDEA 1 / 4 — THE “WIRELESS” MYTH, BUSTED EARLY

“WiFi” is only the last **few metres**.



Phone

))) radio)))



Router

- Your phone talks to your **router** using radio waves.
- That wireless hop is **tiny** — just to the box on your wall.
- After the router? **No more “air”**.

After that, it's **wires**. Lots of them.



Router



City fibre



Undersea cable



Data centre

- Fibre-optic cables run across cities and countries.
- Giant cables on the **ocean floor** connect the continents.
- ~99% of the internet is physical cable — not air, not satellites.

Your data goes to one **exact computer**.

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you type a name...

youtube.com

...which maps to an address (IP)

142.250.195.78

→ a specific server, in a building

- Every server has an **address** — its IP.
- Your request travels to that address...
- ...and the reply comes **back** to your phone.

Milliseconds? It's **light through glass**.

MUMBAI → LONDON

~7,000 km

≈ **120 ms** round trip

- Fibre carries your data as **pulses of light**.
- Light is the **fastest thing** in the universe.
- So even across the planet, it feels instant.

WHY IT MATTERS

Now buffering & lag make sense.

- A server **far away** = more distance = more delay. That's **ping / lag** in games.
- "The cloud" is just **servers in big buildings**, joined by these cables.
- Cut one undersea cable, and a whole region can slow down.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

X MYTH

"The internet is wireless — it floats through the air and satellites."

✓ REALITY

~99% is **physical cable**, including undersea fibre between continents. Wireless (WiFi, mobile) is only the **last hop** to you.

RECAP — 2 THINGS TO REMEMBER

- 1 The internet is a giant web of **physical cables** — wireless is just the last few metres to you.
- 2 Your data races as **light** to a server's address and back — that's the millisecond magic.

The internet = cables + light. 

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Next question: What really happens when you click "Send"? 